Updated State String for Simulation Service, Network Structure 4

chars currently in use in state str:

Used in filepaths, `device_type`, `connection_type`, or our Logs, Reports, Solutions

```
,
;
<alphanumeric>
/
\
[
]
(
)
-
.
\n
```

Potential candidates for state str delimiter:

```
#
$
%
&
!
@
_
+
=
```

Currently considered candidate: `#`

new `sim_state_str` syntax:

```
'<sim_id>#<config>#[<Node_1 str>;<Node_2 str>;]'
```

Note that the Config state str syntax must also use **non-reserved** delimiter(s)

Update to Node-specific State String Syntax

current syntax:

```
<type>,<ip1>.<ip2>.<ip3>.<ip4>,<base_sec_level>,<security_level>,
<risk_level>,<connections_list>

connections list (two entries):
[([<d1_ip1>,<d1_ip2>,<d1_ip3>,<d1_ip4>],<cost>,<bandwidth>,
<connection_type>,<source port number>),([<d2_ip1>,<d2_ip2>,<d2_ip3>,
<d2_ip4>],<cost>,<bandwidth>,<connection_type>,<source port number)]
```

need to **make sure that all of the groups have their own unique syntax** - the program sees it all as one string, so unique delimiters are needed for each **level** of syntax:

Outermost layer (layer 1): Simulation State string ('#')

Outermost layer of the state string should have the following:

- `sim_id`, `config`, `node_list`
  the delimiter for this layer is **#**

```
<sim_id>#
<config>#
[<type>%[<ip1>&<ip2>&<ip3>&<ip4>]%<base_sec_level>%<security_level>%
<risk_level>%[([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)];
<type>%[<ip1>&<ip2>&<ip3>&<ip4>]%<base_sec_level>%<security_level>%
<risk_level>%[([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)]
```

Layer 2: config and node list (';')

Layer two contains `config`'s within the config list, and `node` entries within the node list

Config

`config`'s are going to be user-defined or user-selected configuration entries. See Config state str syntax

the delimiter for this layer is ;

```
[<type%,[<ip1>&<ip2>&<ip3>&<ip4>]%<base_sec_level>%<security_level>%
<risk_level>%[([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)];
<type>%[<ip1>&<ip2>&<ip3>&<ip4>]%<base_sec_level>%<security_level>%
<risk_level>%[([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)]
```

Layer 3: config details and node data ('%')

Layer 3 contains the details and data for each `config` and `node`

the delimiter for this layer is %

```
// THIS IS THE NODE DATA ONLY BELOW. CONFIG SYNTAX IS STILL WORK IN PROGRESS
<type>%
[<ip1>&<ip2>&<ip3>&<ip4>]%
<base_sec_level>%
<security_level>%
<risk_level>%
[([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)]
```

Layer 4: `Node` connections, `Node` IP address ('&')

Layer 4 contains the `Node` `ip_address` octets, and individual `connections` for that `Node`

The delimiter for this layer is &

```
<ip1>&
<ip2>&
<ip3>&
<ip4>

([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)
```

Layer 5: `Connection` details ('@')

Layer 5 contains the details for each `connection` entry

The delimiter for this layer is @

```
[<d1_ip1>=<d1_ip2>=<d1_ip3>=<d1_ip4>]@
<cost>@
<bandwidth>@
<connection_type>@
<source port number>
```

Layer 6: `Connection` ip details ('=')

Layer 6 contains the individual octets for each `connection`'s destination IP address

The delimiter for this layer is =

```
<d1_ip1>=
<d1_ip2>=
<d1_ip3>=
<d1_ip4>
```

FULLY UPDATED SYNTAX

```
<sim_id>#<config>#[<type>%[<ip1>&<ip2>&<ip3>&<ip4>]%<base_sec_level>%
<security_level>%<risk_level>%[([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)];
<type>%[<ip1>&<ip2>&<ip3>&<ip4>]%<base_sec_level>%<security_level>%
<risk_level>%[([<d1_ip1>=<d1_ip2>=<d1_ip3>=
<d1_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)&
([<d2_ip1>=<d2_ip2>=<d2_ip3>=
<d2_ip4>]@<cost>@<bandwidth>@<connection_type>@<source port number>)]
```